

1st Year Exam: Numerical Analysis, April, 2025.  
Do **4** (four) problems.

1. Consider the function  $g(x) = e^{-x}$ .

(a) Prove that  $g$  is a contraction on  $G = [\ln 1.1, \ln 3]$ .

(b) Prove that  $g$  maps  $G = [\ln 1.1, \ln 3]$  into  $G = [\ln 1.1, \ln 3]$ .

(c) Prove that  $x_{k+1} = g(x_k)$  converges to a unique fixed point  $z \in G = [\ln 1.1, \ln 3]$  for any initial value  $x_0 \in G = [\ln 1.1, \ln 3]$ .

2. Based on  $u_1(x) = 1, u_2(x) = x, u_3(x) = x^2$ , use Gram-Schmidt orthogonalization process to compute the three polynomials  $w_1(x), w_2(x), w_3(x)$  which are orthonormal on the interval  $[0, 1]$  with respect to the inner product  $(f, g) = \int_0^1 f(x)g(x)dx$ . Using these polynomials, find the best approximation in  $P^2[0, 1]$  for  $f(x) = x^{\frac{1}{2}}$ .

3. Consider the finite difference formula

$$f'(t_j) = \frac{1}{12h} [f(t_j - 2h) - 8f(t_j - h) + 8f(t_j + h) - f(t_j + 2h)] + O(h^4).$$

(a) Derive this formula by using Taylor's theorem.

(b) Derive this formula by using Lagrange polynomial representation.

4. Assume the numerical quadrature for  $\hat{f}(\hat{x})$  on  $[0, 1]$  is

$$\hat{J}(\hat{f}) = \int_0^1 \hat{f}(\hat{x})d\hat{x} \approx \hat{Q}(\hat{f}) = \sum_{j=0}^m \hat{\alpha}_j \hat{f}(\hat{x}_j).$$

Derive the numerical quadrature of  $J(f) = \int_a^b f(x) dx$ .

5. Consider numerically solving the initial value problem

$$y'(t) = f(t, y), \quad 0 < t \leq t_f, \quad \text{with } y(0) = \eta.$$

Assume  $f$  is sufficiently differentiable and let  $h$  denote the step size. Show that all convergent members of the family of methods

$$y_{n+2} + (\theta - 2)y_{n+1} + (1 - \theta)y_n = \frac{1}{4}h[(6 + \theta)f_{n+2} + 3(\theta - 2)f_n],$$

parameterized by  $\theta$ , are also  $A_0$ -stable.